

# JANHAVI GAIKWAD

Systems Engineer | Systems Architecture | MBSE | Digital Twin | ASEP | GPA : 4.1/4.3  
Ithaca, NY | janhavigaikwad17@gmail.com | [Portfolio](#)

## EDUCATION

### Cornell University : MEng in Systems Engineering

Ithaca, NY | May 2026

- Certification: INCOSE Associate Systems Engineering Professional (ASEP)
- Coursework: Model-Based Systems Engineering · Systems Architecture · System Analysis & Behavior Optimization · Model-Based Estimation · Feedback Control Systems · Digital Twin · Data Science & AI for Systems Engineering · Project Management

### Priyadarshini College of Engineering : B.E. in Aeronautical Engineering

Nagpur, India | Nov 2020

- Coursework: Computational Fluid Dynamics · Aerodynamics · Propulsion · Structures · C Programming

## WORK EXPERIENCE

### Honeywell Technology Solutions

Bengaluru, India

Mechanical Design Engineer : Systems Performance & Operability Methods

Jan 2024 – Jul 2025

- Delivered 300+ hours of engineering productivity gains by designing, implementing, and validating OD engine performance algorithms in C++ (OOP), incorporating gas property corrections, trap efficiency, and scalar logic across fan, compressor, turbine, and combustor modules, enabling rapid system-level trade studies.
- Reduced simulation setup time by ~5 min per run by enabling UI-based access and visualization of 3D performance map data, improving modeling efficiency and system integration usability.
- Maintained feature accuracy within 0.05% tolerance by validating new turbojet and turbofan features in NexSys against FAST legacy models, ensuring interface compatibility and reliable system-level performance assessment.
- Supported cross-functional innovation teams by adapting engine performance models across configurations, contributing analytical insight to efficiency-focused system-level trade and requirements studies.
- Developed a proof-of-concept ML system for electric UAM route optimization, enabling early-stage system studies projected to reduce fuel consumption by ~10%, demonstrating digital reasoning applied to operational system behavior.
- Developed a proof-of-concept ML model to predict gas turbine surge and stall, contributing to operability risk assessment and supporting requirements derivation for next-generation propulsion systems.
- Elicited and refined feature requirements through direct stakeholder engagement in a global agile environment — translating engineering needs into scoped tickets, estimating delivery effort, and maintaining >95% say-do ratio across sprint cycles.
- Enhanced team-wide knowledge retention by authoring monthly Kaizen documentation capturing micro-optimizations, modeling insights, and interface shortcuts, building a continuously evolving system reference.

### Entuple Technologies Ltd.

Pune, India

CFD Application Engineer (ANSYS Fluent)

Aug 2022 – Jan 2024

- Enabled ANSYS Fluent adoption across aerospace, automotive, and research clients by providing training, application support, and simulation best practices for 50+ customers, improving model accuracy and simulation reliability.
- Improved simulation proficiency by ~40% for 500+ engineers through 20+ technical training programs, achieving an average trainee satisfaction score of 4.8/5 and reducing modeling errors by ~25%.
- Accelerated customer adoption by developing 10+ POC CFD models covering multiphase flows, pollutant dispersion, chemical reaction modeling, and moving/dynamic meshes, enabling early validation of complex system-level simulations.
- Served as the primary technical interface for 50+ clients, conducting structured needs analysis to diagnose simulation integration gaps, translating client pain points into actionable solution requirements, and validating outcomes against their operational criteria.
- Resolved complex CFD challenges for strategic clients through model setup guidance and simulation validation support, achieving ~20% faster model turnaround and higher confidence in simulation results.
- Delivered a keynote at ICPHD'23 (IIT Guwahati) showcasing ANSYS simulation tools for green energy and sustainability-focused engineering, reaching 100+ attendees.

### Hindustan Aeronautics Ltd. (HAL)

Nashik, India

Apprentice Trainee : Aerodynamics & Systems Analysis

Aug 2021 – Aug 2022

- Achieved <1% error in lift and drag predictions under operational conditions by implementing a MATLAB/Simulink-based system identification framework on flight data recorder (FDR) data, modeling aircraft dynamic behavior for design validation.
- Evaluated meshing approaches for complex aircraft geometries through a comparative study across multiple tools and techniques, optimizing CFD workflow for solution accuracy and computational efficiency.
- Enabled comparative aerodynamic system analysis by performing CFD simulations of aircraft and airfoil geometries using ANSYS and CFD++, supporting design validation and performance assessment.

- Enhanced CFD workflow efficiency by developing an optimized FORTRAN preprocessing algorithm, reducing computation time for 600,000-element meshes from 50–60 min to 1–2 min, a 30–50x speedup enabling higher-fidelity iterative system analysis.

## ACADEMIC PROJECTS

---

### **MBSE: Solar-Aware Route Optimizer — Cameo** [\[Link\]](#)

- Translated 6 prioritized customer needs into measurable system requirements using Customer Affinity Analysis and Customer Value Proposition (CVP), defining performance thresholds and design priorities.
- Established end-to-end operational clarity by developing Operational Description Templates (ODTs) and context diagrams to formally define subsystem interactions, event sequences, constraints, and data flows across all use cases.
- Controlled interface complexity by constructing an Interface Matrix capturing structural dependencies, data exchanges, and design constraints between all subsystems, enabling traceability of interface changes throughout the lifecycle.
- Reduced requirements ambiguity by formalizing system behavior through SysML use case, activity, sequence, state, and parametric diagrams capturing user interactions, internal logic, and measurable performance constraints.
- Improved design decision efficiency by 20% through structured trade and risk analysis using a 32-tool MBSE toolchain including decision matrices, AHP, QFD, FMEA, fault tree analysis, and risk priority number tracking across the full system lifecycle.
- Ensured lifecycle traceability by deriving originating and derived requirements mapped to behavioral test plans and test methodologies via SysML requirements diagrams and a Verification Cross Reference Matrix (VCRM).

### **Digital Twin: Solar-Aware Route Optimizer — Unity/C#** *(MBSE-governed implementation of the Cameo architecture above)*

- Developed an MBSE-governed digital twin of Manhattan in Unity by importing six city sections from government open datasets, cleaning and integrating building and road data into a unified, interactive city-scale simulation model.
- Developed a C# algorithm to construct a city-wide road network, computing segment lengths, endpoints, and solar shadow parameters to enable physics-accurate, solar-aware route evaluation within the digital twin environment.
- Implementing a multi-algorithm route optimizer using Visibility Graph, Dijkstra, and EJOB optimization to find solar-optimized paths between locations, enabling pre-execution validation of operational strategies defined in the MBSE architecture.

### **Systems Architecture: Next-Gen Aviation Aircraft** [\[Link\]](#)

- Led end-to-end system architecture development for a hybrid-electric general aviation aircraft, translating stakeholder needs into 20 traceable program goals and allocating them to subsystem requirements.
- Developed SysML-based functional, operational, and physical architectures including state machines, use cases, BDD/IBD models, allocated architectures, and a 19-component Design Structure Matrix (DSM) to systematically manage mechanical, electrical, informational, and fluid coupling across subsystem interfaces prior to detailed design.
- Modeled system-level reliability using series-parallel reliability block diagrams (RBDs), quantifying propulsion reliability across all 184,000+ architecture variants as a formal evaluation metric, enabling reliability-driven architecture selection alongside cost, certification schedule, and durability.
- Designed a constraint-driven architecture exploration framework enumerating 184,000+ feasible hybrid-electric aircraft configurations using full-factorial enumeration and multi-objective Pareto optimization, replacing ad hoc selection with evidence-based architecture down-selection.
- Applied NSGA-II genetic algorithm (26-bit chromosome encoding, population 200, 200 generations) to converge on Pareto-optimal architecture families without exhaustive search, recovering dominant decisions (propulsion topology, motor count, FCC redundancy) consistent with full tradespace analysis.

### **Frequency-Domain SISO Turbine Control — MATLAB** [\[Link\]](#)

- Designed and validated a frequency-domain SISO controller in MATLAB for an academic turbine system, minimizing inlet flow disturbance effects by adjusting spool speed while maintaining reference tracking and system stability margins.

### **System Analysis & Behavior Optimization: Airport Preboarding** [\[Link\]](#)

- Developed a Python-based discrete-event simulation of airport preboarding operations modeling passenger flow across check-in, transit, security, and boarding stages to assess system throughput and congestion behavior.
- Performed Monte Carlo simulation and statistical analysis to evaluate queue behavior, throughput, and system stability under varying arrival conditions, validating simulation credibility against operational assumptions.
- Conducted a staffing and capacity trade study for security screening, identifying resource configurations that reduced operational cost while maintaining target throughput, demonstrating cost-performance systems thinking.

### 3D Urban Air Mobility (UAM) Path Planning [\[Link\]](#)

- Designed and implemented 3D UAM path planning frameworks over a Hong Kong city section with no-fly zones, evaluating Visibility Graph + Dijkstra and buffered 3D A\* routing architectures.
- Delivered a structured trade study highlighting trade-offs between path optimality, computational cost, and scalability to support architecture selection for urban-scale autonomous routing systems.

### Supply Chain Intelligence Dashboard — SupplyMindAI [\[Link\]](#)

- Architected an integrated data-to-decision pipeline (Shiny + Supabase + Posit Connect) providing real-time visibility into in-transit shipments, hub geography, and escalation paths for critical delays.
- Developed an agentic AI layer with tool-calling routing natural-language queries to analytics modules (monitoring, historical optimization, simulation), with explainable outputs and confidence-aware reasoning per shipment.
- Designed simulation-backed decision support for evaluating delay reduction, capacity, and routing strategies using cost-performance trade studies, enabling pre-execution validation of operational levers.

### Smart Charge — EV Grid Carbon Optimization [\[Link\]](#)

- Developed a real-time EV charging decision-support system converting live grid carbon intensity data into optimized low-emission charging schedules, integrating external APIs, LLM-based reasoning, and calendar systems into an end-to-end operational pipeline.

## LEADERSHIP & PROFESSIONAL ENGAGEMENT

---

### INCOSE Student Chapter : Vice President

Cornell University | Sep 2025 – May 2026

- Improved usability of XRP Robotics documentation and enhanced campus visibility of INCOSE through organized events promoting systems engineering principles and student engagement.

### Eagle Student Forum : Technical Head

Nagpur, India | Aug 2018 – Jul 2019

- Managed technical logistics for departmental programs involving 200 students and coordinated Vaimanika intercollegiate project competition across 30 projects and 150 students.

## SKILLS

---

**Certifications:** INCOSE Associate Systems Engineering Professional (ASEP)

**Systems Engineering:** Systems Architecture · Requirements Engineering · Verification & Validation (V&V) · DoDAF · MBSE (SysML, Cameo) · Interface Matrix · ODT · DSM · Tradespace & Pareto Analysis · Multi-Objective Optimization · Reliability Modeling (RBD) · QFD · FMEA · FFBD · IDEFO · CVP · VCRM · Discrete-Event Simulation · Interface Control · System Integration · Project Management (WBS, EVM, Risk)

**Digital Twin & Modeling:** Digital Twin Development (Unity/C#) · OD Gas-Turbine Performance Modeling · System Identification · Simulink · Physics-Based Simulation · CFD (ANSYS Fluent, CFD++) · ANSYS Meshing · NexSys · FAST

**Controls & Estimation:** Feedback Control (Time & Frequency Domain) · State-Space Design · Extended Kalman Filtering · State Estimation · SISO/MIMO Control Design

**Programming & Data:** Python · MATLAB · C++ · C# · C · FORTRAN

**AI & Analytics:** Agentic AI · LLM Integration · Tool-Calling · Multi-Agent Workflows · Machine Learning (POC) · Monte Carlo Simulation · Statistical Analysis · REST APIs

**Tools & Platforms:** Cameo (MagicDraw) · ANSYS Fluent · Simulink · Visual Studio · CATIA V5 · STK · ProjectLibre · Posit Connect · Shiny for Python · Supabase · GitHub